

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**

Enterprise Java

Module - I

**Practical – III**

|                             |                                   |
|-----------------------------|-----------------------------------|
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**3. Implement the Servlet IO and File applications.**

**a. Create a Servlet application to upload and download a file.**

**CODE:-**

**Index.html:**

```
<!DOCTYPE html>
<html>
<head>
<title>File Upload</title>
</head>
<body>
<h1>File Upload Application</h1>
<form action="FileUploadServlet" method="post" enctype="multipart/form-data">
Select file to upload: <input type="file" name="file" required>
<br><br>
<input type="submit" value="Upload File">
</form>
<h2>Download Links</h2>
<!-- Link to download files, ensure the path is correct -->
<a href="FileDownloadServlet?filename= Image of IOT Prac1,2.pdf">
file</a><br><br><br><br>
</body>
</html>
```

**FileUploadServlet.java:**

```
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;
import javax.servlet.ServletException;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.Part;
@WebServlet("/FileUploadServlet")
```

```

@MultipartConfig
public class FileUploadServlet extends HttpServlet {
private static final String UPLOAD_DIR = "uploads";
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
String uploadPath = getServletContext().getRealPath("") + File.separator + UPLOAD_DIR;
File uploadDir = new File(uploadPath);
if (!uploadDir.exists()) {
uploadDir.mkdir();
}
Part filePart = request.getPart("file");
String fileName = filePart.getSubmittedFileName();
File file = new File(uploadPath + File.separator + fileName);
try (InputStream inputStream = filePart.getInputStream();
OutputStream outputStream = new FileOutputStream(file)) {
byte[] buffer = new byte[4096];
int bytesRead;
while ((bytesRead = inputStream.read(buffer)) != -1) {
outputStream.write(buffer, 0, bytesRead); } }
response.getWriter().println("File uploaded successfully Priti Yadav.");
} }

```

#### **FileDownloadServlet.java:**

```

import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.OutputStream;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/FileDownloadServlet")
public class FileDownloadServlet extends HttpServlet {
private static final String UPLOAD_DIR = "uploads";
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
String fileName = request.getParameter("filename");
if (fileName == null) {
response.getWriter().println("File name is missing.");
return; }
String filePath = getServletContext().getRealPath("") + File.separator + UPLOAD_DIR +
File.separator + fileName;
File file = new File(filePath);
if (!file.exists()) {
response.getWriter().println("File not found.");
return; }
response.setContentType("application/octet-stream");
response.setHeader("Content-Disposition", "attachment; filename=\"" + fileName +
"\");
try (FileInputStream inputStream = new FileInputStream(file);
OutputStream outputStream = response.getOutputStream()) {

```

```
byte[] buffer = new byte[4096];
int bytesRead;
while ((bytesRead = inputStream.read(buffer)) != -1) {
    outputStream.write(buffer, 0, bytesRead);
}
}
```

#### web.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE glassfish-web-app PUBLIC "-//GlassFish.org//DTD GlassFish Application Server
3.1 Servlet 3.0//EN" "http://glassfish.org/dtds/glassfish-web-app_3_0-1.dtd">
<glassfish-web-app error-url="">
<context-root>/FileUploadanDownload</context-root>
<class-loader delegate="true"/>
<jsp-config>
<property name="keepgenerated" value="true">
<description>Keep a copy of the generated servlet class' java code.</description>
</property>
</jsp-config>
</glassfish-web-app>
```

**OUTPUT:-**

## File Upload Application

Select file to upload:  Image of IOT Prac1,2.pdf

### Download Links

[Download the uploaded file](#)

The screenshot shows a web application interface for file uploads. On the left, there's a sidebar with navigation links: Desktop, Downloads (selected), Documents, and Pictures. The main area displays a table of uploaded files:

| File Name            | Upload Date         | Browser              | Size |
|----------------------|---------------------|----------------------|------|
| Image of IOT Prac1,2 | 02-07-2026 07:14 PM | Microsoft Edge PD... | 2    |
| EJ Prac1 T054        | 02-07-2026 06:59 PM | Microsoft Edge PD... | 3    |
| SE S053              | 02-07-2026 06:58 PM | Microsoft Word D...  | 2,6  |
| Mern Stack SRMS T054 | 29-06-2026 08:52 PM | Microsoft Edge PD... | 2    |

Below the table, there's a file selection dialog. The 'File name' field contains 'Image of IOT Prac1,2'. The 'File type' dropdown is set to 'All files'. At the bottom, there are three buttons: 'Upload from mobile', 'Open' (highlighted), and 'Cancel'.

File uploaded successfully.

### Downloads



Image of IOT Prac1,2 (2).pdf  
[Open file](#)

**b. Develop Simple Servlet Question Answer Application using Database.****CODE:-**

```
CREATE DATABASE qadb;
```

```
USE qadb;
```

```
CREATE TABLE quiz
```

```
(  
  qno varchar(5) PRIMARY KEY,  
  question varchar(200),  
  op1 varchar(100),  
  op2 varchar(100),  
  op3 varchar(100),  
  op4 varchar(100),  
  ans varchar(100)  
);
```

```
INSERT INTO quiz VALUES
```

```
('001','Which keyword is used to inherit a class in  
Java?','implements','extends','inherit','using','extends');
```

```
INSERT INTO quiz VALUES
```

```
('002','Which method is the entry point of a Java  
program?','start()','main()','run()','init()','main()');
```

```
INSERT INTO quiz VALUES
```

```
('003','Which package contains Scanner  
class?','java.io','java.util','java.lang','java.sql','java.util');
```

```
INSERT INTO quiz VALUES
```

```
('004','Which is not a primitive data type?','int','float','String','char','String');
```

```
INSERT INTO quiz VALUES
```

```
('005','Which exception occurs when a number is divided by  
zero?','IOException','NullPointerException','ArithmeticException','ClassNotFou  
ndException','ArithmeticException');
```

**Index.html:**

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Java Quiz</title>
```

```
</head>
```

```
<body bgcolor="lightyellow">
```

```
<center>
```

```
<h1>Java Quiz Application</h1>
```

```
<h3>Click below to start the quiz</h3>
```

```
<a href="QuizServlet">START QUIZ</a>
```

```
</center>
```

```
</body>
```

```
</html>
```

```
QuizServlet.java
```

```
package mypack;
```

```
import java.io.*;
```

```
import java.sql.*;
```

```
import javax.servlet.*;
```

```
import javax.servlet.http.*;
```

```
public class QuizServlet extends HttpServlet
```

```

{
public void doGet(HttpServletRequest request,
HttpServletResponse response)
throws ServletException, IOException
{
response.setContentType("text/html");
PrintWriter out = response.getWriter();
out.println("<html>");
out.println("<body bgcolor='lightblue'>");
out.println("<center>");
out.println("<h1>Java Quiz</h1>");
out.println("</center>");
out.println("<form action='ShowResult'>");
try
{
Class.forName("com.mysql.cj.jdbc.Driver");
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/qadb",
"root",
"root");
Statement st = con.createStatement();
ResultSet rs = st.executeQuery("select * from quiz");
int i = 1;
while(rs.next())
{
out.println("<hr>");
out.println("<h3>Q"+i+" . "
+rs.getString("question")+
"</h3>");
out.println("<input type='radio' name='"+i+"'
value='"+rs.getString("op1")+"'>" +rs.getString("op1")+"<br>");
out.println("<input type='radio' name='"+i+"'
value='"+rs.getString("op2")+"'>" +rs.getString("op2")+"<br>");
out.println("<input type='radio' name='"+i+"'
value='"+rs.getString("op3")+"'>" +rs.getString("op3")+"<br>");
out.println("<input type='radio' name='"+i+"'
value='"+rs.getString("op4")+"'>" +rs.getString("op4")+"<br>");
i++;
}
con.close();
}
catch(Exception e)
{
out.println(e);
}
out.println("<br>");
out.println("<input type='submit' value='Submit Quiz'>");
out.println("<input type='reset' value='Reset'>");
out.println("</form>");
out.println("</body>");
out.println("</html>");
}

```

```

}
ShowResult.java
package mypack;
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class ShowResult extends HttpServlet
{
    public void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException
    {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html>");
        out.println("<body bgcolor='lightgreen'>");
        out.println("<center>");
        out.println("<h1>Quiz Result</h1>");
        out.println("</center>");
        try
        {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/qadb",
                "root",

                "root");
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select ans from quiz");
            int score = 0;
            int i = 1;
            while(rs.next())
            {
                String correct = rs.getString("ans");
                String user = request.getParameter(""+i);
                if(correct.equals(user))
                {
                    score++;
                    out.println("<h3>Q"+i+" : Correct</h3>");
                }
                else
                {
                    out.println("<h3>Q"+i+" : Incorrect</h3>");
                }
                i++;
            }
            out.println("<hr>");
            out.println("<h2>Your Score : "+score+" / "+(i-1)+"</h2>");
            con.close();
        }
        catch(Exception e)

```

```
{
out.println(e);
}
out.println("<br><a href='index.html'>Take Quiz Again</a>");
out.println("</body>");
out.println("</html>");
}
}
```

**web.xml:**

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
version="3.1">
<servlet>
<servlet-name>QuizServlet</servlet-name>
<servlet-class>mypack.QuizServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>QuizServlet</servlet-name>
<url-pattern>/QuizServlet</url-pattern>
</servlet-mapping>
<servlet>
<servlet-name>ShowResult</servlet-name>
<servlet-class>mypack.ShowResult</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>ShowResult</servlet-name>
<url-pattern>/ShowResult</url-pattern>
</servlet-mapping>
</web-app>
```

**OUTPUT:-**

## Java Quiz Application

Click below to start the quiz

[START QUIZ](#)

**Q1. Which keyword is used to inherit a class in Java?**

- ☐ implements
- ☒ extends
- ☐ inherit
- ☐ using

**Q2. Which method is the entry point of a Java program?**

- ☐ start()
- ☒ main()
- ☐ run()
- ☐ init()

**Q3. Which package contains Scanner class?**

- ☐ java.io
- ☒ java.util
- ☐ java.lang
- ☐ java.sql

**Q4. Which is not a primitive data type?**

- ☐ int
- ☒ float
- ☐ String
- ☐ char

**Q5. Which exception occurs when a number is divided by zero?**

- ☐ IOException
- ☒ NullPointerException
- ☐ ArithmeticException
- ☐ ClassNotFoundException

Submit Quiz

Reset

## Quiz Result

**Q1 : Correct**

**Q2 : Correct**

**Q3 : Correct**

**Q4 : Incorrect**

**Q5 : Incorrect**

**Your Score : 3 / 5**

[Take Quiz Again](#)

**c. Create simple Servlet application to demonstrate Non-Blocking Read Operation.**

**CODE:-**

**index.html:**

```
<!DOCTYPE html>
<html>
<head>
<title>Non-Blocking Practical</title>
</head>
```



```

<body>
<form action="NBServ" method="post">
<h3> Hey Khushi Click Start Button to Start Non-Blocking Read!!!</h3>
<input type="submit" value="Start NB IO Read">
</form>
<div>TO DO write content</div>
</body>
</html>

```

#### **MyReadListener.java:**

```

package EJPracts;
import java.io.IOException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.AsyncContext;
import javax.servlet.ReadListener;
import javax.servlet.ServletInputStream;
public class MyReadListener implements ReadListener {
private ServletInputStream iStream = null;
private AsyncContext cnt = null;
public MyReadListener(ServletInputStream in, AsyncContext ac) {
this.iStream = in;
this.cnt = ac;}
@Override
public void onDataAvailable() {
try {
StringBuilder sb = new StringBuilder();
int len;
byte[] b = new byte[1024];
while (iStream.isReady() && (len = iStream.read(b)) != -1) {
String mydata = new String(b, 0, len);
sb.append(mydata);
} } catch (IOException ex) {
Logger.getLogger(MyReadListener.class.getName()).log(Level.SEVERE, null, ex);
} }
@Override
public void onAllDataRead() {
System.out.println("Called onAllDataRead()");
cnt.complete();
}
@Override
public void onError(Throwable t) {
Logger.getLogger(MyReadListener.class.getName()).log(Level.SEVERE, null, t);
cnt.complete();
} }

```

#### **NBServ.java:**

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet("/NBServ")
public class NBServ extends HttpServlet {
protected void processRequest(HttpServletRequest request, HttpServletResponse response)

```

```

        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    String filepath = "/WEB-INF/student.txt";
    ServletContext cnt = getServletContext();
    InputStream is = cnt.getResourceAsStream(filepath);
    PrintWriter out = response.getWriter();
    if (is == null) {
        out.println("<h2>File NOT FOUND!</h2>");
        return;
    }
    BufferedReader br = new BufferedReader(new InputStreamReader(is));
    String line;
    out.println("<html><body>");
    out.println("<h2>Student Details:</h2>");

    while ((line = br.readLine()) != null) {
        out.println(line + "<br>");
    }
    out.println("</body></html>");
    br.close();
    is.close();
}

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
}

ReadNBServ.java:
import EJPracts.MyReadListener;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.AsyncContext;
import javax.servlet.ServletException;
import javax.servlet.ServletInputStream;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "ReadNBServ", urlPatterns = {"/ReadNBServ"}, asyncSupported = true)
public class ReadNBServ extends HttpServlet {
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
        out.println("<!DOCTYPE html>");
        out.println("<html>");

```

```
out.println("<head>");
out.println("<title>Servlet ReadNBServ</title>");
out.println("</head>");
out.println("<body>");
AsyncContext cnt = request.startAsync();
ServletInputStream iStream = request.getInputStream();
iStream.setReadListener(new MyReadListener(iStream, cnt));
out.println("</body>");
out.println("</html>");}}
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
processRequest(request, response);}
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
processRequest(request, response);}
@Override
public String getServletInfo() {
return "Short description";}
}
```

**Details.txt file:**

Khushi Patel  
T032  
BSCIT  
TYIT  
MVLU

**OUTPUT:-**

**Hey Khushi Click Start Button to Start Non-Blocking Read!!!**

Start NB IO Read

TO DO write content

**Student Details:**

Khushi Patel  
T032  
BSCIT  
TYIT  
MVLU